Muhammad Sarmad Saleem

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Professional Summary

Software Engineer with experience in full-stack development, backend systems, and AI-integrated applications. Skilled in building scalable web applications using modern frameworks, designing efficient APIs, and developing robust system architectures using Object-Oriented Programming (OOP) principles and software engineering best practices. Proficient in data structures, algorithms, database design, and real-time systems development. Experienced in delivering production-ready solutions with a strong focus on performance, reliability, scalability, and user experience.

Technical Skills

Programming Languages: Java, Python, JavaScript, C, C++
Frontend Development: ReactJS, NextJS, HTML, CSS
Backend Development: NodeJS, ExpressJS, Django, Flask, FastAPI, REST APIs
Databases: MongoDB, MySQL, PostgreSQL
Additional Technologies: HuggingFace, LangChain, LangGraph, RAG Pipelines, Pytorch, Tensorflow, Pandas, Numpy, OpenCV
Real-Time Systems: WebSockets, WebRTC,
Tools & Version Control: Git, GitHub, Docker

Experience

Fully Funded DAAD Research Intern — Jul 2025 – Aug 2025
Robotics Research Lab, RPTU Kaiserslautern, Germany

- Conducted deep learning research on glacier extent and elevation prediction under the CASREW Project.
- Worked on satellite and climate data-based modeling and evaluation pipelines.
- Presented research findings to international researchers and faculty.

Software Engineer (Part-Time) — Sep 2024 – Oct 2024
Search O' Pal

- Developed a job recommendation system matching candidate skills with job roles to improve hiring pipeline efficiency.
- Built backend APIs and integrated automated NLP-based resume parsing system.
- Improved onboarding workflows and reduced manual processing time.

Software Engineer Intern — Apr 2024 – Aug 2024
Search O' Pal

- Built scalable assessment system supporting MCQs, short answers, and evaluation workflows.
- Implemented real-time video interviews using WebRTC and Jitsi integration.
- Developed real-time messaging system using WebSockets for multi-user communication.

Education

Bachelor of Science in Computer Science — 2022 – 2026
National University of Sciences & Technology (NUST), Islamabad — CGPA: 3.41 / 4.00

FSc Pre-Engineering — 2020 – 2022
Government College, Lahore — Marks: 1061 / 1100

Projects

Digital Logic Design Simulator

- Built a Java-based interactive simulator for designing and visualizing digital logic circuits.
- Implemented real-time circuit evaluation engine for combinational logic operations.
- Enhanced learning experience by enabling hands-on simulation of digital systems.

GlacioVision: Glacier Monitoring & Prediction Platform

- Developed an AI-powered web platform for predicting glacier extent and elevation using satellite imagery and climate data.

- Built an interactive dashboard to visualize historical glacier changes and model-based forecasts for decision support.
- Enabled real-time access to predictive analytics, improving usability of complex geospatial insights for non-technical users.

Assessment & Interview Platform

- Developed a full-stack recruitment platform using Django and React for online assessments and candidate evaluation.
- Implemented REST APIs for exam management, scoring, and user workflows.
- Integrated WebRTC and Jitsi for scalable real-time video interviews supporting concurrent users.

Faryaad-e-Rehmat: Charity Donation Platform

- Built a full-stack MERN application for donation management, case submission, and verification workflows.
- Developed backend services for secure data handling, donation tracking, and beneficiary management.
- Improved transparency and efficiency in charitable donation distribution through automated workflows.

Bank Chatbot: RAG + Fine-Tuned LLM

- Built a RAG-based chatbot for banking support using FastAPI and Streamlit.
- Used FAISS for semantic retrieval from multi-format documents (PDF, DOCX, JSON, Excel).
- Fine-tuned LLM to generate structured, domain-specific responses.
- Designed API-based system with real-time querying and incremental indexing.

LawBot: Legal Document Retrieval System

- Built a web-based legal document search system using RAG pipelines and semantic embeddings.
- Developed backend APIs for querying structured legal datasets efficiently.
- Improved legal information access through fast and accurate document retrieval.

Achievements

DAAD Fully Funded Research Internship

Selected for international research internship at RPTU Kaiserslautern, Germany.

Research Paper Acceptance at IGARSS

Accepted research work on glacier prediction using deep learning.

Leadership & Activities

- Founder & President – Faryaad-e-Rehmat
- Team Lead – ACM NUST
- Director HR – IEEE NUST